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Magnetic Stratigraphy of the Ordovician in the Lower Reach of the Kotuy River: the Age of the Bysy-Yuryakh Stratum and the Rate of Geomagnetic Reversals on the Eve of the Superchron

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Abstract—Until recently, the existing data prevented the geophysicists from accurately dating the Bysy-Yuryakh stratum, which outcrops in the middle reach of the Kotuy River, constraining the time of its formation to a wide interval from the end of the Late Cambrian to the beginning of the Silurian. The obtained paleomagnetic data unambiguously correlate the Bysy-Yuryakh stratum to the Nyaian regional stage and constrain its formation, at least a considerable part of it, by the Tremadocian. This result perfectly agrees with the data on the Bysy-Yuryakh conodonts studied in this work and yields a spectacular example of the successful application of paleomagnetic studies in solving important tasks of stratigraphy and, correspondingly, petroleum geology. Within the Bysy-Yuryakh stratum, we revealed a large normal-polarity interval corresponding to the long (>1 Ma) period when the geomagnetic reversals were absent. This result, in combination with the data for the Tremadocian and Middle–Upper Cambrian sequences of the other regions, indicates that (1) the rate of occurrence of the geomagnetic reversals on the eve of the Ordovician Moyero superchron of reversed polarity was at most one reversal per Ma; (2) the superchron does not switch on instantaneously but is preceded by a certain gradual change in the operation conditions of the dynamo mechanism which, inter alia, manifests itself by the reduction of the frequency of geomagnetic reversals with the approach of the superchron. This finding supports the views according to which a process preparing the establishment of the superchrons takes place at the core–mantle boundary.

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INTRODUCTION

One of the most topical tasks in the studies of the evolution of the geomagnetic field lies in obtaining the information about the time changes in the rate of geomagnetic reversals with the approach of a superchron. The gradual reduction in the reversal rate is interpreted as an indication that somewhere in the Earth's interior, probably at the core–mantle boundary, a certain process takes place leading to the gradual change in the generation conditions of the magnetic field in the outer core (Courtillot and Besse, 1987; Larson and Olson, 1991; Courtillot and Olson, 2007; Olson et al., 2013; Biggin et al., 2012). In contrast, a sharp shutdown of the reversals is understood as a manifestation of the stochastic nonlinear nature of the geodynamo and as evidence that a process preparing the arrival of the superchron is absent (Hulot and Gallet, 2003; Aubert et al., 2009). The fact of a protracted reduction in the frequency of reversals on approaching (from the past) the most explored—Cretaceous—superchron had not been doubted (e.g., see (McFadden and Merril, 1996)) until this conclusion was challenged and interpreted as an artifact by (Gallet and Hulot, 1997). (Strictly speaking, they did not fully reject the possibility of the existence of a process that prepared the superchron. They only constrained the duration of this process by an order of magnitude shorter compared to the previous researchers).

In the ensuing controversy (McFadden and Merril, 2000), a certain intermediate period was chosen by the work (Constabe, 2000), where it was shown that the existing data were insufficient for selecting, with due strictness, which of the alternative standpoints was correct. It has become clear that to resolve this ques-

tion, it is necessary to accumulate new experimental data describing the changes in the polarity of the geomagnetic field as a function of time with the approach of the other superchrons—the Cretaceous/Permian Kiama superchron (Opdyke and Channel, 1996) and the Early Ordovician Moyero superchron, which was fairly recently established (Pavlov and Gallet, 2005). In this work, we present the new magnetostratigraphic data which, in combination with the previous results, support the opinion about the gradual reduction in the frequency of the reversals, in particular, with the approach of the Early Ordovician superchron.

Ideally, such data should be derived by the magnetostratigraphic investigation of the reliably dated sedimentary sections that have long been accumulated before the occurrence of superchrons. Unfortunately, in reality these sections are very scarce, some parts do not contain the primary paleomagnetic information, other parts contain significant hiatuses in the sedimentation, some parts are poorly exposed, etc. Most of the known reference Late Cambrian—Early Ordovician (deposited before the Moyero superchron) sections of North Eurasia have been studied already; however, since there are only a few such sections, any new information about the polarity of the geomagnetic pole during the corresponding time period is highly important for data interpretation and for substantiating this interpretation. This is the type of information presented in this paper.

The section of the lower reach of the Kotuy River had not been magnetostratigraphically explored until recently. This appears surprising because the section is comparatively easily accessible, largely composed of red rocks favorable for paleomagnetic studies, has a fairly large thickness, and is perfectly well exposed. However, this situation is probably accounted by the fact that the geological literature barely contains information that would allow reasonably accurate (at the regional stage level) stratigraphic positioning of the rocks composing the section.

Our studies yielded the data that allowed us to significantly refine the age of the Kotuy section.

STUDY OBJECT

The studied section outcrops in the lower reach of the Kotuy River in the region of the Kazyl-Kaya stow and stretches along the right coast of the river for more than 2 km (Fig. 1). According to the lithological signs, the section includes two strata. The lower Tompo stratum is mainly represented by green clayey dolomites and in places with marl interbeds; the upper Bysy-Yuryakh stratum erosionally overlays the Tompo stratum and is composed of red dolomite marls and clayey dolomites with gypsum intercalations. The rocks are generally have monocline bedding and gently dip west-northwest at an angle of 2°–8°.

The Tompo stratum concordantly covers the underlying Upper Cambrian sediments and has a thickness of ~60 m. The thickness of the Bysy-Yuryakh layers in this section is ~115 m. Southwest of the study region, the Bysy-Yuryakh layers with a stratigraphic gap are overlain by the Lower Silurian formations of the Llandovery Group.

Judging by the published data (*Gosudarstvennaya*..., 1962), the faunal characterization of these formations is extremely scarce and limited to the presence of *Dendrograptus* sp. graptolites (the Upper Cambrian–Silurian stratigraphic range) and *Dimogchillina* sp. ostracodes (Ordovician–Silurian).

The Ordovician age of the Tompo and Bysy-Yuryakh strata on the Kotuy River is determined by their position in the section between the faunally characterized rocks of the Upper Cambrian and Lower Silurian. We note however that here the Upper Cambrian age of the underlying rocks requires additional validation.

Thus, the faunal data that have been published before the present study determine the age of the rocks composing the Kotuy section as Ordovician at best.

The Ordovician age of the formations composing the studied section is suggested by the results of their comparison with the faunally characterized coeval sediments of the other regions. In particular, in the section of the Moyero River a few hundred km south of the Kotuy section, which has a similar stratigraphic position, the Ordovician guide faunal remains have been found (*Gosudarstvennaya*..., 1962).

We also note that the Bysy-Yuryakh stratum of the Maymecha River, which outcrops about 100 km southwest of the discussed section, is overlain by the Ust-Kuranakh Formation, which is fairly well characterized by Ugrian fauna (Kanygin et al., 2007). This probably led some authors to attribute the Bysy-Yuryakh stratum in some cases to the Mansian, Loparian, and Nyaian horizons and in other cases simply to the Nyaian horizons (Kanygin et al., 2007; 2013) (Fig. 1). However, the time correlation of the distant strata which is, in fact, exceptionally based on the lithological signs alone is not strict; therefore, the age constraints existing for the Bysy-Yuryakh stratum in the Maymecha valley are, generally speaking, not necessarily valid for the Bysy-Yuryakh stratum of the Kotuy River.

The Tompo stratum is poorly dated by fauna in all the sections where it was studied. Hence, its attribution to particular horizons of the Lower Ordovician of Upper Cambrian (*Resheniya*..., 1983) in different stratigraphic schemes should be considered as rather speculative.

PALEOMAGNETISM

The Ordovician stratum of the Kazyl-Kaya stow was sampled by us from the bottom upwards along the

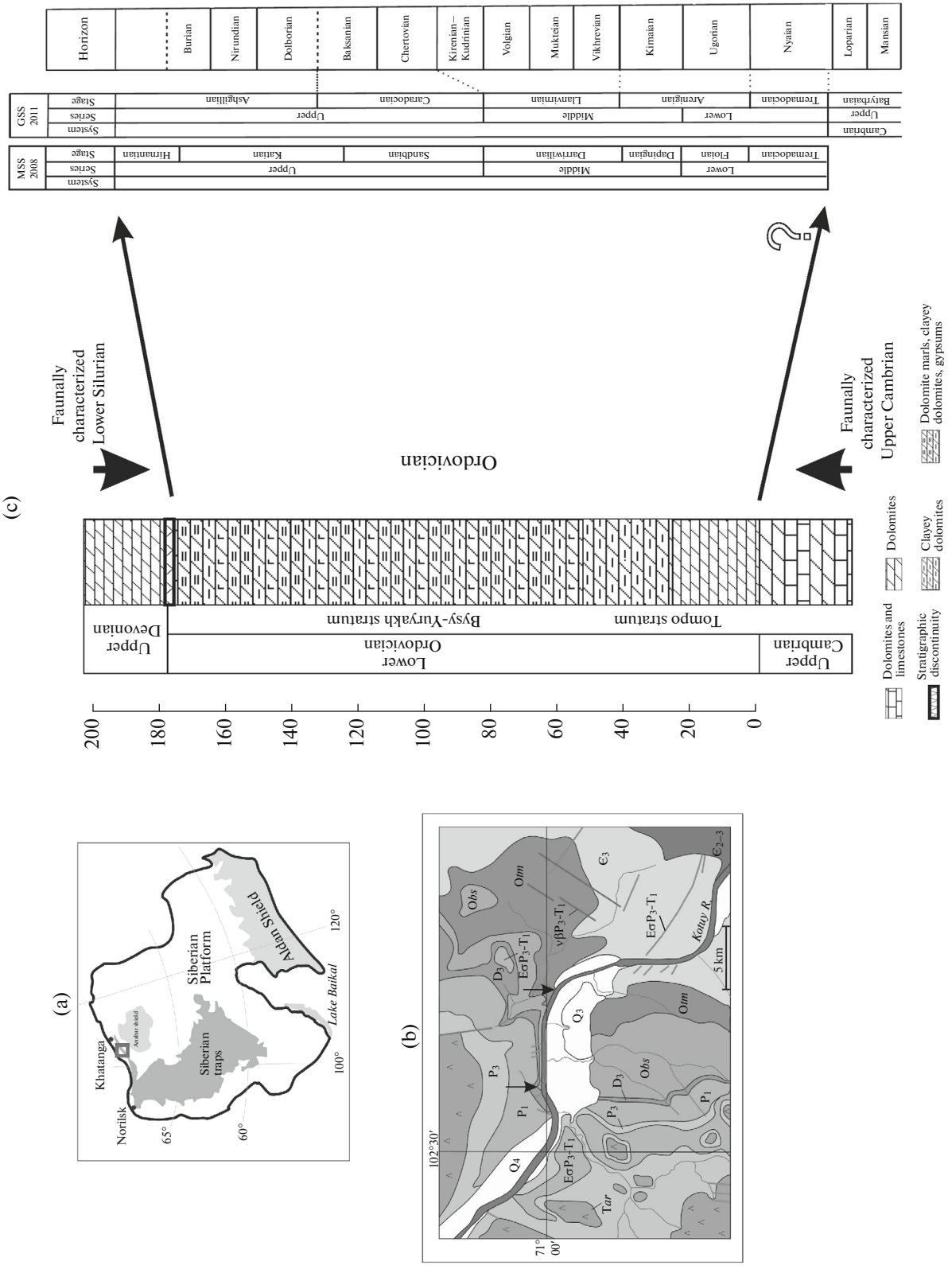


Fig. 1. (a) Geographic position of study region; (b) geologic scheme of study region. Black arrows mark interval of outcrop of studied section on right coast of Kotui River; (c) stratigraphic position and lithology of studied section.

Table 1. Magnetization components

Component, number of samples	Geographic coordinate system				Stratigraphic coordinate system			
	D (°)	I (°)	K	α95 (°)	D (°)	I (°)	K	α95 (°)
OR component $N = 28$	149.2	−43.3	19.4	6.4	146.5	−42.3	19.9	6.3
TR+ component $N = 12$	15.5	86.8	135.2	3.7	310.1	85.7	119.0	4.0
TR− component $N = 7$	286.6	−61.6	27.5	11.7	286.3	−65.8	55.9	8.1
Pole of OR component: Plat = 143.5°; Plong = −39.7°; dp/dm = 4.8°/7.8°; A95 = 6.1°								

D—declination, I—inclination, K—concentration, α95—radius of 95%-confidence circle.

section with an interval of 0.5–2 m. The sampling points were more-or-less uniformly distributed across the section with the only significant gap corresponding to the practically completely unexposed interval slightly thicker than 20 m in the upper part of the Tompo stratum. A total of 150 oriented blocks were yielded from which 2-cm cubic specimens were sawn out.

The laboratory paleomagnetic studies conducted by the standard procedure have shown that only one-third of the samples contain the ancient paleomagnetic record. In the other samples, stable magnetization is either absent or is represented by a recent magnetization component.

Thermal demagnetization resolves two ancient magnetization components (Table 1, Fig. 2). The Permian–Triassic trap (TR) component is related to magnetite, as suggested by the values of the maximal deblocking temperatures (Figs. 2a, 2b). The Ordovician (OR) component is carried by hematite, which again follows from the maximal deblocking temperatures (Fig. 2c). The age of these components is determined quite confidently since the directions of these components for Siberia are well known due to the numerous paleomagnetic studies previously carried out by our colleagues and us (Khramov et al., 1982; Rodionov and Gurevich, 2010; Gallet and Pavlov, 1996; Pavlov and Gallet, 1998; Pavlov et al., 2007; 2008). Practically in all the samples carrying ancient magnetization, these components occur separately. The positions of the samples carrying these magnetization components are shown in Fig. 3.

Since the age of the considered rocks is Ordovician, the trap component is likely to be secondary, metachronous. The trap remagnetization of the Paleozoic rocks of Siberia is an ordinary phenomenon and has often been described in the literature (e.g., (Gallet and Pavlov, 1996; Pavlov and Gallet, 1998)). In this case, however, this remagnetization has a specificity that merits separate discussion. This specificity consists of the fact that the trap component is represented by the directions of both the normal and reversed polarity. Although the possibility of the coexistence of the metachronous components having very close ages and different polarities in one section is accepted by many authors, in our practice, this is the first case when it is manifested with perfect clearness and definiteness. To the best of our knowledge, similar unambiguous examples have not been described in the Russian scientific literature.

We note the following properties of the revealed trap component of the magnetization.

(1) The mean directions calculated for the vectors of normal (TR+) and reversed polarity (TR−) are clearly nonantipodal (Fig. 2d).

(2) These mean directions are very close to the paleomagnetic directions of the trap flows of the Arydzhangsky Formation studied by us previously and located relatively close to the considered section (Fig. 2f). This can be considered as an indication that it is the magmatic events associated with the eruption of these lava flows that have been the source of the discussed remagnetization.

(3) The (sub)components TR+ and TR− are localized within different parts of the section and are characterized by a different quality of the paleomagnetic record (Figs. 2a, 2b, 3).

(4) The high concentration calculated for the magnetization vectors of the trap component (at least, for the TR+ group (Table 1)) testify to the fairly fast formation of this component (which took less time than required for averaging out the secular variations).

(5) There are signs suggesting that the TR− and TR+ remagnetizations occurred before and after the dislocation of the rocks of the studied section, respectively. This observation requires additional studies for their validation. If the suggested correlation is confirmed, it would be logical to link these dislocations to the trap eruption process.

In any case, the presence of a bipolar, definitely metachronous component in the section calls for caution in applying the third Van der Voo's criterion (van der Voo, 1993) for substantiating the primary nature of the magnetization. This raises the following question: how would it be possible to reveal the metachronous

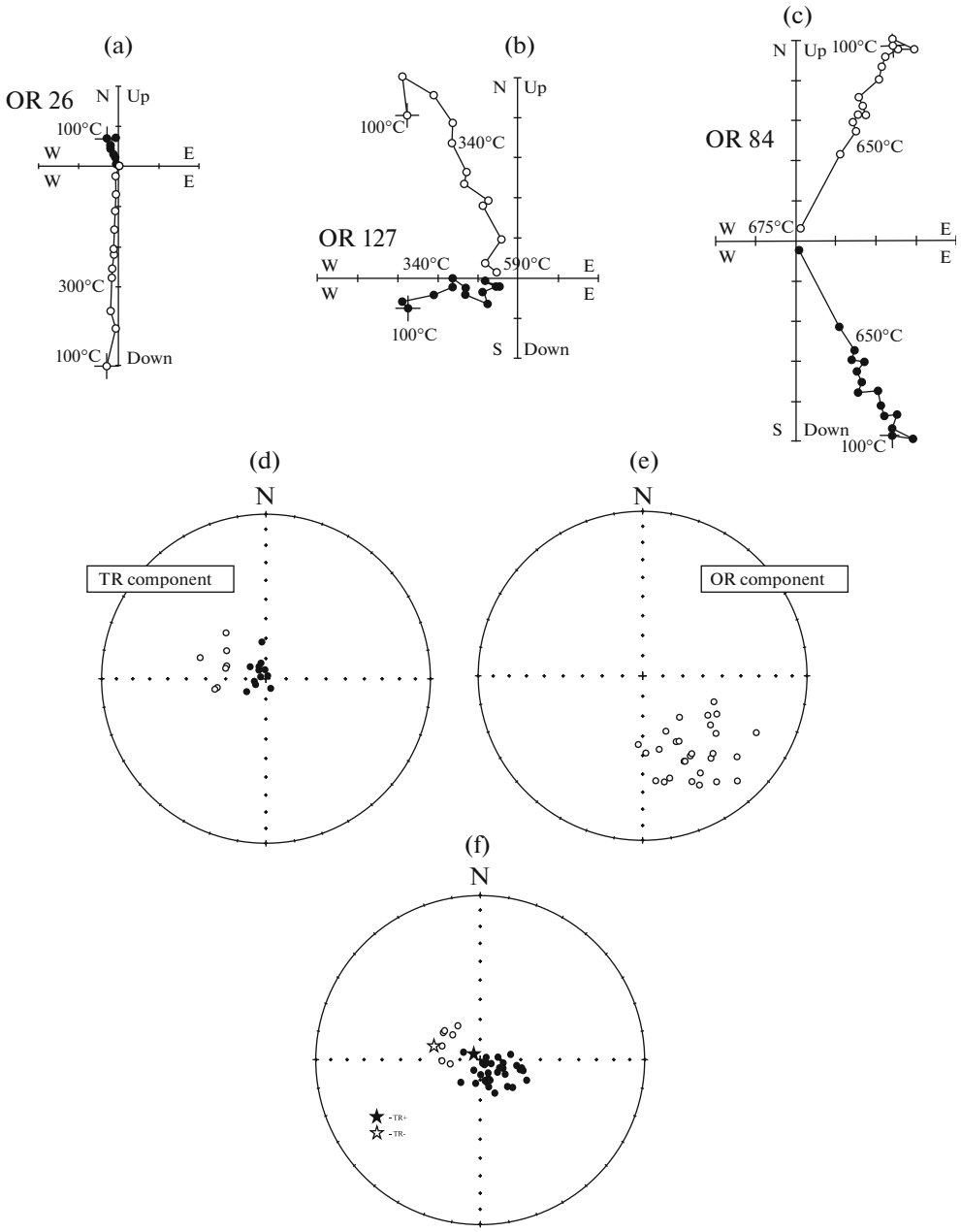


Fig. 2. Zijderveld plots (a, b, c) illustrating presence of Ordovician (OR) and trap (TR+ and TR–) magnetization components, and stereograms illustrating distribution directions of trap (TR) (d) and Ordovician (OR) (e) magnetization components; (f) comparison of the mean directions calculated for the trap component (stars), with the directions determined earlier for the trap volcanic flows in the region of the mouth of the Medvezh'ya River (Pavlov et al., 2011).

nature of the bipolar ancient magnetization component in the absence of a priori information?

The noted nonantipodality of the mean directions with normal and reversed polarity may arise in rapid remagnetization episodes. However, incomplete antipodality can also be caused by other factors, e.g., incomplete separation of the magnetization components during the demagnetization and the presence of nondipole components in the averaged field.

The significant difference in the properties of the components with normal and reversed polarity and in the properties of the rocks carrying these magnetizations should be considered an annoying feature pointing to the probable metachronous nature of the considered component. However, the very fact of this difference is insufficient for asserting its secondary origin, at least because of the fact that each layer and any group of layers in any sedimentary section are, to a certain extent, peculiar by their characteristics.

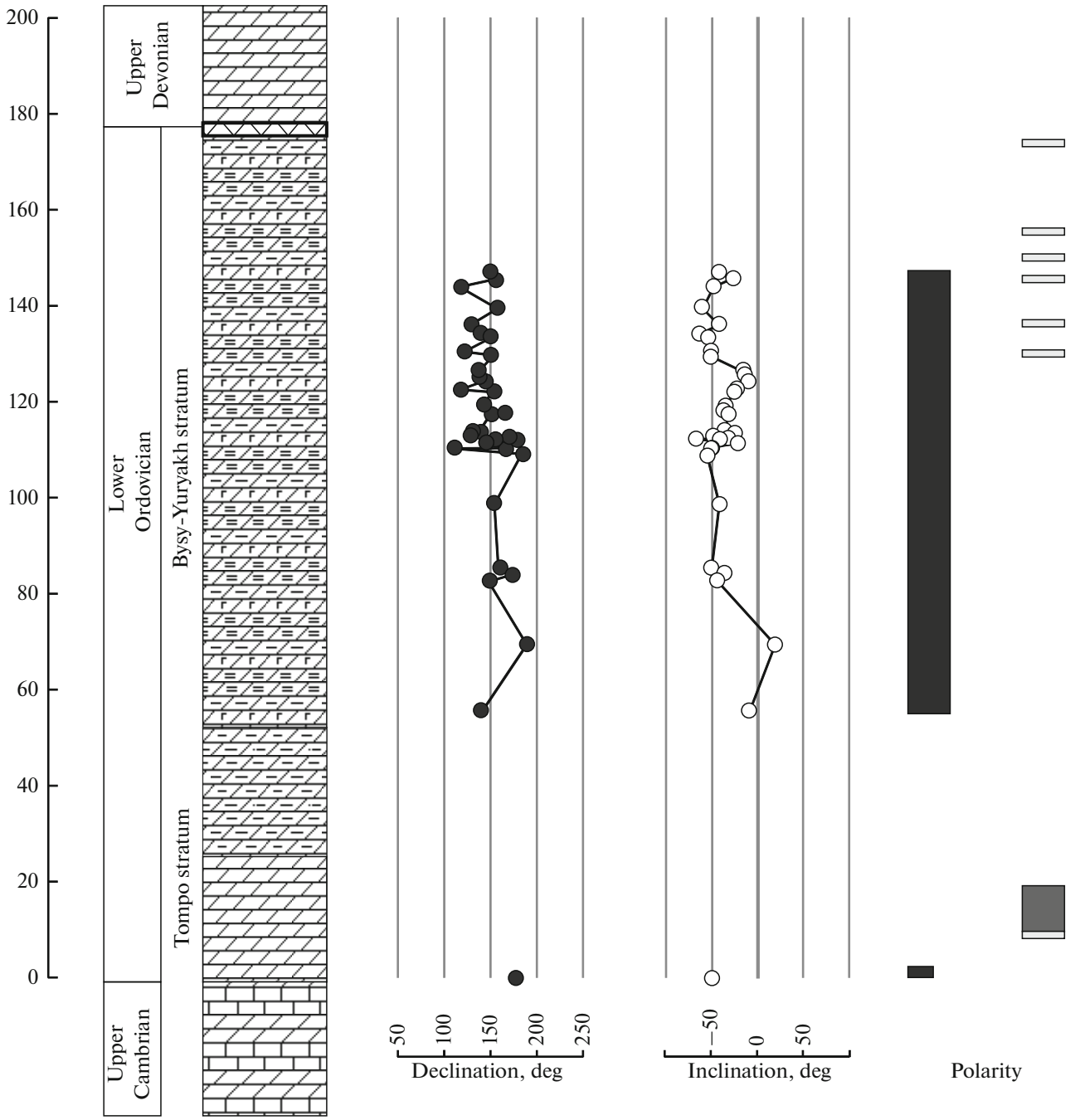


Fig. 3. Magnetostratigraphy of Kazyl-Kaya section. Polarities: black—normal polarity of Ordovician component; white (gray)—reversed (normal) polarity of trap component.

In the case of revealing the described specific features of the bipolar components, it is perhaps reasonable to conduct the in-depth mineralogical–optical, electron microscopy, and other studies of the magnetization-carrying minerals, in the hope that their morphological peculiarities or specific compositional features will help substantiate the primary or secondary nature of the magnetization. We note, however, that these studies, even sufficiently detailed ones, rarely yield unambiguous results. Therefore, the question formulated above remains unsolved.

Since the Ordovician trend of the paleomagnetic poles for the Siberian Platform is well known, the Ordovician magnetization component revealed in the section can be used for a more accurate dating of the latter. Except for one sample from the very bottom of the section, the Ordovician component is only present in the rocks of the Bysy-Yuryakh stratum; therefore,

the age estimation from the paleomagnetic data as described below, strictly speaking, can only be referred to the upper 2/3 of the section of the Kazyl-Kaya stow.

The paleomagnetic pole calculated from the OR component perfectly agrees with the trend of the Ordovician paleomagnetic poles from the Siberian Platform and falls between the Tremadocian (Nyaian regional stage[1]) and Arenigian poles (the Ugorian and Kimaian regional stages). Here, considering the position of the pole, we may rather speak of the Nyaian and, less probably, of the Ugorian age of the Bysy-Yuryakh pole (Table 2, Fig. 4a).

This conclusion is a major step in dating the Kotuy Ordovician because it significantly narrows the probable age limits for the Bysy-Yuryakh stratum. However, our results allow us to take another step to more accurately constrain the age of these rocks.

Numerous data indicate that since the beginning of the Arenigian (Ugorian age on the Siberian regional scale), the geomagnetic chron of reversed polarity (Moyero superchron) had been established in the Earth and probably lasted at least 15 Ma (Pavlov and Gallet, 2005). However, the Bysy-Yuryakh rocks are only magnetized in the field of normal polarity. This unambiguously rejects the Ugorian age of the Bysy-Yuryakh stratum (Fig. 4b) and gives grounds to assert that it was formed in the Nyaian (Tremadocian).

BIOSTRATIGRAPHIC DATA

The paleomagnetic data obtained by us are simple, unequivocal, and confidently constrain the stratigraphic position of the considered stratum with the maximal possible accuracy (Fig. 4). Meanwhile, in the geological and geophysical studies, it is extremely desirable that even the most reliable conclusions be supported by data obtained by independent methods.

Considering this, we attempted to isolate and investigate the conodonts from the material that remained after sawing the sampled blocks and preparing the paleomagnetic specimens. From a part of this material, we isolated solitary conodonts *Glyptoconus* cf. *G. quadraplicatus* (Branson et Mehl) and *Glyptoconus* sp. *G.* cf. (Fig. 5).

G. quadraplicatus is represented by the Sc and Sa elements and not exactly identified. *Glyptoconus* sp. is distinguished by its lathe size and acute frontal edge of the base. Based on such limited material, confident identification up to a species of the revealed forms is impossible. Nevertheless, the specimens of this genus are widespread in the Ugorian–lower part of the Kimaian horizons in the Siberian Platform and in the interval of the topmost Tremdocian–lower part of the Floian Stage in North America. Finding elements of only the *Glyptoconus* genus suggests the Tremadocian age of the hosting rocks rather than a younger age where these forms are ubiquitous in significantly more diverse conodont complexes.

Thus, the new data on conodonts perfectly agree with the age determination for the upper part of the Bysy-Yuryakh stratum based on the paleomagnetic data.

Despite the mentioned speculative character of the correlation of the Bysy-Yuryakh strata in the Kotuy and Maymecha valleys, we believe that the faunal complexes for Maymecha can be used in support of the age constraints directly obtained from the Kotuy section.

In this context, the attempt to characterize the Bysy-Yuryakh Formation in the typical region of the formation on the left coast of the Maymecha River 2 km downstream from the Bysy-Yuryakh River mouth, which was made during the recent geological surveying works on refining the R-47 sheet (1 : 1000000), is undoubted interest. Conodonts were absent in the predominant part of the samples; several conodont elements of *Proconodontus notchpeakensis* (Miller), *Proconodontus posterocostatus* Miller and *Cordylodus* sp. were found only in a bulk sample capturing 2 m of the section of the lower part of the formation (Fig. 5). Finding these species, which are characteristic of the lower part of the Nyaian horizon, supports our conclusion based on the paleomagnetic data for the Kotuy section. Interestingly, during the same survey on the Maymecha River, in the Ust-Kuranakh Formation overlying the Bysy-Yuryakh one, conodonts were not found; however, their occurrence corresponding to the Ugorian horizon, together with rare brachiopods and gastropods, has been noted for this region in previous studies (Kanygin et al., 2007).

DISCUSSION

An important feature of the magnetostratigraphic record in the Kazyl-Kaya section consists of the relatively large thickness of the revealed interval of normal polarity measuring about 90 m. Conversion from the thickness of the stratigraphic interval of the section to the estimate of the duration of its accumulation is challenging. This can be done based on the rates of sedimentation known for the carbonate platforms, which are frequently a few cm per ka and a few dozen meters per Ma (Bosscher and Schlager, 1993). In this case, the formation of the considered interval should take at least 2–3 Ma.

Another probable approach consists of comparing sections that have closer lithological characteristics and were accumulated in similar tectonic settings (epicontinental basins). Several such examples can be found in the neighboring regions of the Siberian Platform, e.g., the Ordovician section of the Moyero River (the southern Anabar region) and the Kulumbe River

[1] Although we support the opinion that the lower boundary of the Nyaya horizon runs somewhat below the Ordovician/Cambrian boundary (Tolmacheva and Abaimova, 2009), hereinafter, as suggested (Kanygin et al., 2013), the Nyaya horizon is for simplicity correlated to the Tremadocian in full.

Table 2. Paleomagnetic poles of Angara–Anabar block of Siberian Platform

No.	Object, region, coordinates	Plat (°)	Plong (°)	A95 (°)	Data source
	Ordovician				
	Kotuy River, 70.0° N; 102.7° E	–39.7	143.5	6.1	This work
	Tremadocian (Nyaian horizon)				
1	Moyero River, 67.5° N; 104° E	–40.3	137.5	6.9	(Gallet and Pavlov, 1996)
2	Kulumbe River, 68° N; 88.8 ° E	–35.2	127.2	4.1	(Pavlov and Gallet, 1998)
3	Kulumbe River, 68° N; 88.8° E	–36	138	4.7	(Rodionov and Gurevich, 2010)
Tremadocian MEAN	PLAT = –37.3°, PLONG = 134.2°, A95 = 6.8°				
	Arenigian (Ugorian, Kimaian horizons)				
4	Moyero River, 67.5° N; 104° E	–33.9	151.7	1.9	(Gallet and Pavlov, 1996)
5	Rozhkova River (middle reach of Angara River). 58.5° N; 99.8° E	–36.4	158.2	6.5	(Pavlov et al., 2011)
6	Kulumbe River, 68° N; 88.8° E	–32.1	113	5.3	(Rodionov and Gurevich, 2010)*
Arenigian MEAN	PLAT = –35.2°, PLONG = 154.9°, A95 = 8.5°				
	Llanvirnian (Mukteian and Vikhorevian horizons)				
7	Moyero River, 67.5° N; 104° E	–29.8	156.6	3.1	(Gallet and Pavlov, 1996)
8	Kulumbe River, 68° N; 88.8° E	–30.9	152.7	2.8	(Pavlov and Gallet, 1998)
9	Rozhkova River (middle reach of Angara River), 58.5° N; 99.8° E	–35.2	153.2	3.6	(Pavlov et al., 2012)
Llanvirnian MEAN	PLAT =-31.6°, PLONG = 153.7°, A95 = 4.7°				
	Llandeilo (Volgian and Kirenian–Kudrian horizons)				
10	Moyero River, 67.5° N; λ = 104.0° E	–22.7	157.6	2.8	(Gallet and Pavlov, 1996)
11	Kulumbe River, 68° N; λ = 88.8° E	–24.1	152.4	3.3	(Pavlov et al., 2008)
12	Stolbovaya River, 62.1° N; λ = 92.5° E	–22.0	158.0	4.0	(Pavlov et al, 2008)
Llandeilo MEAN	PLAT=–25.2°, PLONG = 156.2°, A95 = 8.5°				
	Caradocian–Ashgillian (Chertovian, Baksanian, Dolborian horizons)				
13	Rozhkova River (middle reach of Angara River), 58.5° N; 99.8° E	–29.5	140.2	6.4	(Pavlov et al., 2011)
14	Moyero River, 67.5° N; 104.0° E	–13.9	124.1	5.9	(Gallet and Pavlov, 1996)
Caradocian–Ashgillian MEAN	PLAT = –21°, PLONG = 132°, A95 = 9.0°				

* Pole is not used because calculated from direction strongly contaminated by remagnetizing trap component (Rodionov and Gurevich, 2010). Plat and Plong are the latitude and longitude of the paleomagnetic pole, respectively; A95—radius of 95 %-confidence circle.

section (the northeastern regions of the Siberian Platform).

The Arenigian sediments of the Moyero River (the Floian and Dapingian stages in the General stratigraphic scale) do not have visible breaks and are composed of rocks that are similar to the Kotuy rocks and represented by clayey limestones with the alternation of gypsum interbeds (Gallet and Pavlov, 1996). The Arenigian here has a thickness of ~105 m and was accumulated during ~15 Ma (*Geological...*, 2012). The compositions close to the discussed Kotuy sediments are also noted for the Guragir Formation of the Ordovician section of the Kulumbe River, which are attributed to the Llanvirnian in its traditional understanding (Gradstein and Ogg, 1996) and presently correlated to the lower 2/3 of the Darriwilian Stage (Kanygin et al., 2013). These sediments are dominated by clayey dolomites with lenses of anhydrite; their thickness is ~160 m and time of accumulation ~5 Ma (*Geological...*, 2012).

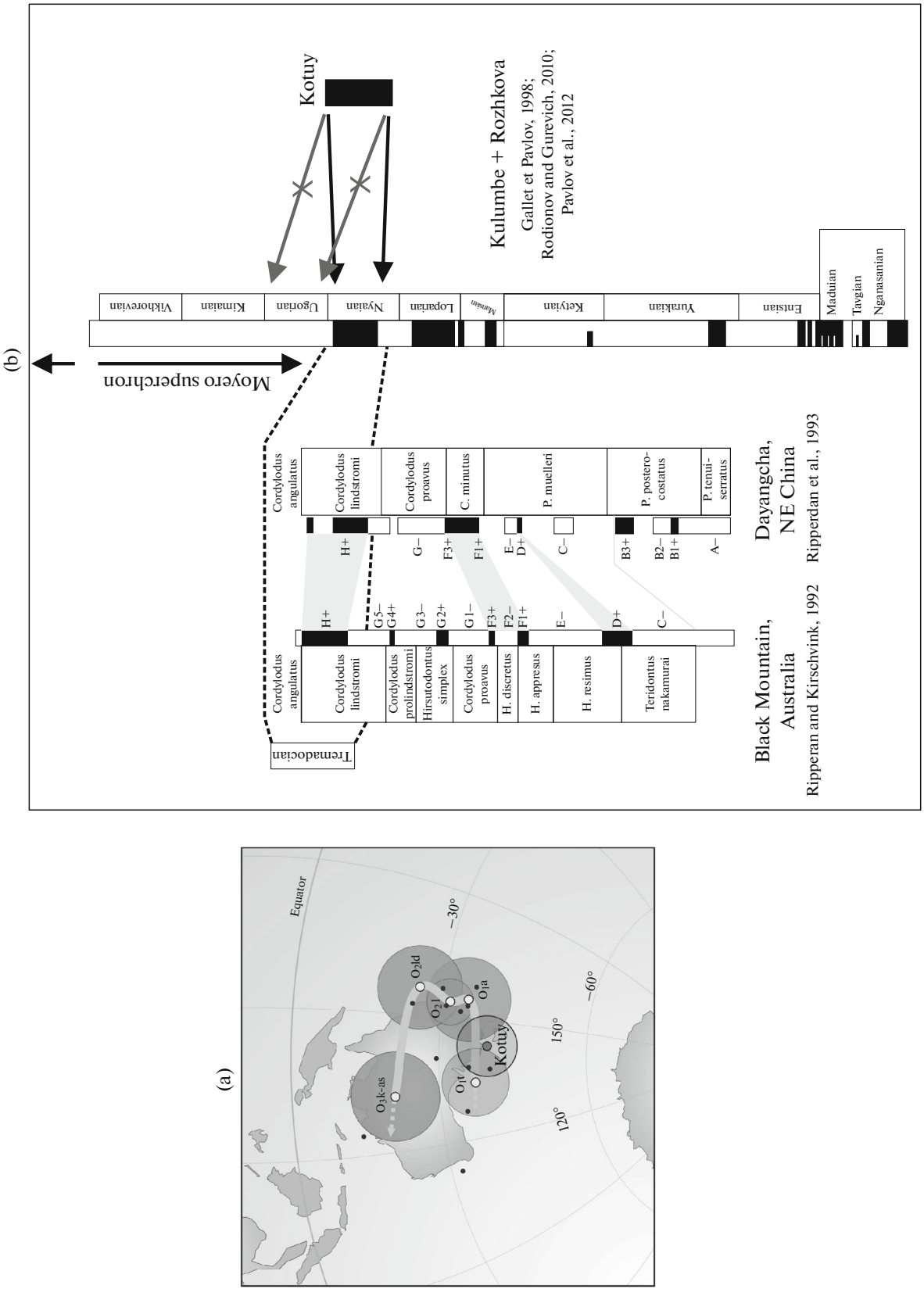


Fig. 4. Paleomagnetic determination of age of OR component: (a) comparison of paleomagnetic pole calculated from OR component with Ordovician paleomagnetic poles for Siberian Platform; (b) comparison of OR component polarity with sequence of magnetic polarity intervals in Early Ordovician.



Fig. 5. Conodonts found in Bysy-Yuryakh stratum of Kotuy River (*1*, *2*, *3*) and Maymecha River (*4*, *5*, *6*). White line segment (100 μm) shows sizes of elements: *1*, *Glyptoconus* sp., Sc element; *2*, *Glyptoconus* cf. *G. quadraplicatus* (Branson et Mehl), Sc element; *3*, *G.* cf. *G. quadraplicatus*, Sa element; *4*, *Euconodontus notchpeakensis* (Miller), Sc element; *5*, *Cordylodus* sp., M element; *6*, *Proconodontus posterocostatus* Miller, Sc element.

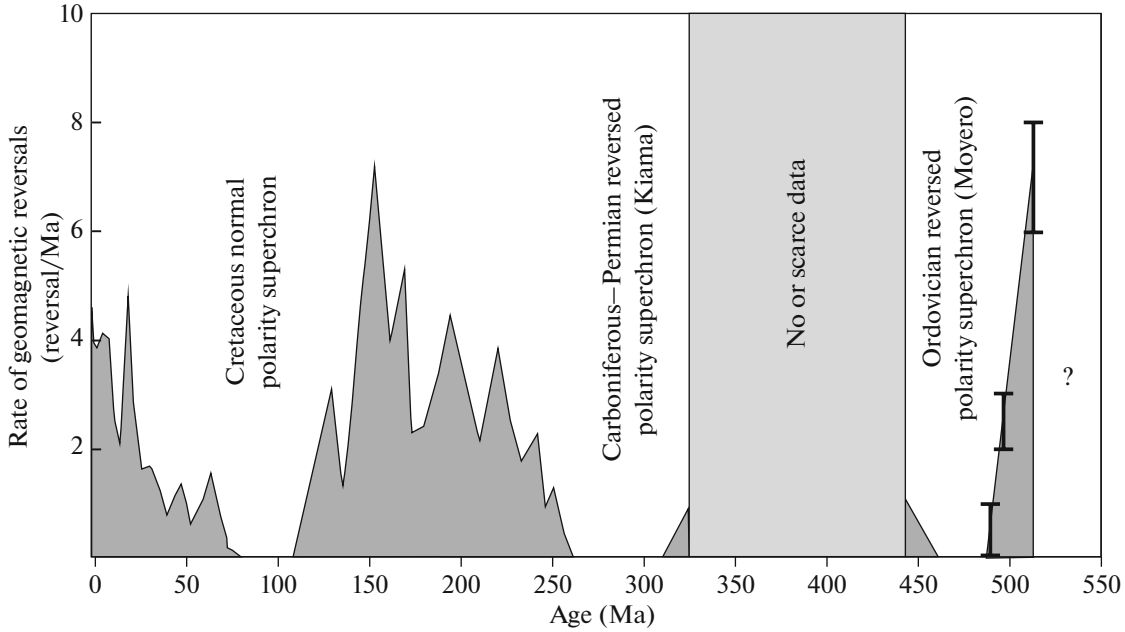


Fig. 6. Change in rate of geomagnetic reversals on eve of beginning of Ordovician reversed polarity superchron (Moyero) (Pavlov and Gallet, 2005).

Reasoning from the data presented above for the Moyero and Kulumbe sections, we again come to the conclusion that the duration of the accumulation of rocks corresponding to the normal polarity interval of the Kazyl-Kaya section cannot be less than a few Ma.

Hence, we have good reasons to believe that the normal polarity interval recorded in the Tremadocian Kotuy section has a duration of at least 1 Ma (and probably much longer). Let us compare this result with the data of the magnetostratigraphic studies of the Tremadocian in the other regions. As of now, the magnetostratigraphy data for the Tremadocian are known for three sections in the world: the Black Mountain in Austalia, Dayang Cha in northeastern China, and Kulumbe in the northwestern part of the Siberian Platform (Ripperdan and Kirschvink, 1992; Ripperdan et al., 1993; Yang et al, 2002; Pavlov and Gallet, 1998). In the Dayangcha and Black Mountain

sections, we see 3–4 magnetic polarity intervals. In both sections there is an interval of normal polarity which has a sufficiently large thickness.

In the Tangchan section, up to five magnetic polarity intervals are confidently established (probably fewer if we take into account the uncertain position of the lower boundary of the Tremadocian in this region), and three intervals are established in the Kulumbe section. Thus, the existing data indicate that in the Tremadocian, during an interval of about 8 Ma (Cohen et al., 2013), 2 to 4 geomagnetic reversals took place.

Hence, the rate of reversals on the eve of the Moyero superchron was significantly lower than in the Middle Cambrian and at the beginning of the Late Cambrian (Fig. 6). This result can be considered as evidence that the superchron is not abruptly switched on but is preceded by a certain gradual change in the operation conditions of the dynamo mechanism which, inter alia, manifests itself by a reduction in the rate of the geomagnetic reversals on the approach of a superchron. These changes are typically linked to the redistribution of the heat flow at the core/mantle boundary which, in turn, can be caused by the evolution of the D" layer, mantle avalanches, and various manifestations of the mantle convection (Courtillot and Besse, 1987; Larson and Olson, 1991; Courtillot and Olson, 2007; Olson et al., 2013; Biggin et al., 2012). The selection of a particular mechanism accounting for the influence of the mantle processes on the geodynamo's operation significantly depends on our knowledge of the character of the variations in the rate of the geomagnetic reversals close to the commencement of superchrons. Primarily we need to understand how gradually these changes occur and which characteristic times describe these changes. For solving this problem for the Ordovician Moyero superchron, we need significantly more detailed information about the structure of the magnetic polarity of the field in the Middle and Late Cambrian than is available as of now. The solution of this problem should become the focus of future studies.

CONCLUSIONS

1. The age of the Bysy-Yuryakh stratum of the Ordovician in the middle reach of the Kotuy River is unambiguously determined by the obtained paleomagnetioc data as Nyaian (Tremadocian). This conclusion is supported by the occurrence of conodonts.

2. The rocks of the studied section contain the bipolar metachronous magnetization component, which was formed due to the intrusion of the Permian–Triassic traps extensively present in the region. In the absence of a priori information, this component could have been easily misinterpreted as primary. The possibility of diagnosing the metachronous nature of the bipolar magnetization components in the absence of a priori information remains unclear.

3. In the section of the Bysy-Yuryakh stratum, a large interval of normal polarity is revealed. This interval corresponds to a long (>1 Ma) period of the absence of geomagnetic reversals. This result, in combination with the data for the Tremadocian sections of the other regions, indicates that the rate of the geomagnetic reversals on the eve of the Moyero Ordovician superchron of reversed polarity was at most (and perhaps much lower than) one reversal per Ma.

4. In the context of the data for the rate of geomagnetic reversals in the Middle and at the beginning of the Late Cambrian, the obtained result can be interpreted as indicating that the superchron is not abruptly switched on but is preceded by a certain gradual change in the operating conditions of the geodynamo which, inter alia, manifests itself by the reduction in the rate of reversals with the approach of the superchron. This result supports the notions according to which the process preparing the occurrence of the superchrons takes place at the core–mantle boundary. The characteristic times of this process can be determined by the detailed magnetostratigraphic studies of the time intervals preceding superchrons.

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REFERENCES

Aubert, J., Labrosse, S., and Poitou, C., Modelling the palaeo-evolution of the geodynamo, *Geophys. J. Int.*, 2009, vol. 179, pp. 1414–1428.

Biggin, A.J., Steinberger, B., Aubert, J., Suttie, N., Holme, R., Torsvik, T.H., van der Meer, D.G., and van Hinsbergen, D.J.J., Possible links between long-term geomagnetic variations and whole-mantle convection processes, *Nat. Geosci.*, 2012, vol. 5, pp. 526–533.

Bosscher, H. and Schlager, W., Accumulation rates of carbonate platforms, *J. Geol.*, 1993, vol. 101, pp. 345–355.

Cohen, K.M., Finney, S.C., Gibbard, P.L., and Fan, J.-X., The ISC International Chronostratigraphic Chart, *Episodes*, 2013, vol. 36, pp. 199–204.

Constable, C., On rates of occurrence of geomagnetic reversals, *Phys. Earth Planet. Inter.*, 2000, vol. 118, pp. 181–193.

Courtillot, V. and Besse, J., Magnetic field reversals, polar wanders, and core-mantle coupling, *Science*, 1987, vol. 237, pp. 1140–1147.

Courtillot, V. and Olson, P., Mantle plumes link magnetic superchrons to Phanerozoic mass depletion events, *Earth Planet. Sci. Lett.*, 2007, vol. 260, pp. 495–504.

Gallet, Y. and Pavlov, V., Magnetostratigraphy of the Moyero River section (northwestern Siberia): constraints on geomagnetic reversal frequency during the Early Palaeozoic, *Geophys. J. Int.*, 1996, vol. 125, no. 1, pp. 95–105.

Gallet, Y. and Hulot, G., Stationary and nonstationary behavior within the geomagnetic polarity time scale, *Geophys. Rev. Lett.*, 1997, vol. 24, no. 15, pp. 1875–1878.

The Geologic Time Scale 2012, vol. 1, Gradstein, F.M., Ogg, J.G., Schmitz, M.D., and Ogg, G.M., Eds., Amsterdam: Elsevier, 2012.

Gosudarstvennaya geologicheskaya karta SSSR. Masshtab 1 : 200000. Seriya Anabarskaya. List: R-48-VII, VIII. Sostavitel': A.I. Ivanov (The State 1 : 200000 Geological Map of the USSR, Anabar Series, Sheet R-48-VII, VIII, Complied by A.I. Ivanov), Savitskii, V.E., Ed., Moscow: Gosgeoltekhizdat, 1962.

Gradstein, F.M. and Ogg, J.G., A Phanerozoic time scale, *Episodes*, 1996, vol. 19, pp. 3–5.

Hulot, G. and Gallet, Y., Do superchrons occur without any palaeomagnetic warning?, *Earth Planet. Sci. Lett.*, 2003, vol. 210, pp. 191–201.

Kanygin, A.V., Yadrenkina, A.G., and Timokhin, A.V., *Stratigrafiya neftegazonosnykh basseinov Sibiri. Ordovik Sibirskoi platformy* (Stratigraphy of the Oil-and-Gas Bearing Basins of Siberia. Ordovician of the Siberian Platform), Novosibirsk: GEO, 2007.

Kanygin, A.V., Yadrenkina, A.G., Timokhin, A.V., Gonta, T.V., and Maslova, O.A., Ordovician biostratigraphic zones of the Siberian platform and the problem of their correlation to the new stages of the International Stratigraphic Scale, in *Regional'naya stratigrafiya pozdnego dokembriya i paleozoya Sibiri: Sb. nauch. tr.* (Regional Stratigraphy of Late Precambrian and Paleozoic of Siberia: Collection of Papers), Novosibirsk: SNIIIGGiMS, 2013, pp. 63–77.

Khramov, A.N., Goncharov, G.I., Komissarova, R.A., Pisarevskii, S.A., Pogarskaya, I.A., Rzhevskii, Yu.S., Rodionov, V.P., and Slautsitais, I.P., *Paleomagnitologiya (Paleomagnetology)*, Leningrad: Nedra, 1982.

Larson, R. and Olson, P., Mantle plumes control magnetic reversal frequency, *Earth Planet. Sci. Lett.*, 1991, vol. 107, pp. 437–447.

McFadden, P.L. and Merrill, R.T., Asymmetry in the reversal rate before and after the Cretaceous normal polarity superchron, *Earth Planet. Sci. Lett.*, 1996, vol. 149, pp. 43–47.

McFadden, P.L. and Merrill, R.T., Evolution of the geomagnetic reversal rate since 160 Ma: is the process continuous?, *J. Geophys. Res.*, 2000, vol. 105, no. B12, pp. 28455–28460.

Olson, P., Deguen, R., Hinnov, L., and Zhong, S., Controls on geomagnetic reversals and core evolution by mantle convection in the Phanerozoic, *Phys. Earth Planet. Inter.*, 2013, vol. 214, pp. 87–103.

Opdyke, N.D. and Channell, J.E.T., *Magnetic Stratigraphy*, San Diego: Academic, 1996.

Pavlov, V. and Gallet, Y., Upper Cambrian to Middle Ordovician magnetostratigraphy from the Kulumbe River section (northwestern Siberia), *Phys. Earth Planet. Inter.*, 1998, vol. 108, pp. 49–59.

Pavlov, V. and Gallet, Y., A third superchron during the Early Paleozoic, *Episodes*, 2005, vol. 28, pp. 78–84.

Pavlov, V., Courtillot, V., Bazhenov, M., and Veselovsky, R., Paleomagnetism of the Siberian traps: new data and a new overall 250 Ma pole for Siberia, *Tectonophysics*, 2007, vol. 443, pp. 72–92.

Pavlov, V., Bachtadse, V., and Mikhailov, V., New Middle Cambrian and Middle Ordovician palaeomagnetic data from Siberia: Llandelian magnetostratigraphy and relative rotation between the Aldan and Anabar-Angara blocks, *Earth Planet. Sci. Lett.*, 2008, vol. 276, no. 3-4, pp. 229–242.

Pavlov, V.E., Fluteau, F., Veselovskiy, R.V., Fetisova, A.M., and Latyshev, A.V., Secular geomagnetic variations and volcanic pulses in the Permian–Triassic traps of the Norilsk and Maimecha–Kotui provinces, *Izv., Phys. Solid Earth*, 2011, no. 5, pp. 35–50.

Pavlov, V.E., Veselovskii, R.V., Shatsillo, A.V., and Gallet, Y., Magnetostratigraphy of the Ordovician Angara/Rozhkova river section: further evidence for the Moyero reversed superchron, *Izv., Phys. Solid Earth*, 2012, vol. 48, no. 4, pp. 297–305.

Resheniya Vsesoyuznogo stratigraficheskogo soveshchaniya po dokembriyu, palozoyu i chetvertichnoi sisteme Srednei Sibiri (Decisions of the All-USSR Stratigraphic Conference on the Precambrian, Paleozoic, and Quaternary of Middle Siberia), part 1, Novosibirsk: SNIIGGiMS, 1983.

Ripperdan, R.L. and Kirschvink, J., Paleomagnetic results from the Cambrian–Ordovician boundary section at Black Mountain, Georgina Basin, western Queensland, Australia, in *Global Perspectives on Ordovician Geology*, Laurie, J.R. and Webby, B.D., Eds., Rotterdam: Balkema, 1992, pp. 93–103.

Ripperdan, R.L., Magaritz, M., and Kirschvink, J., Carbon isotope and magnetic polarity evidence for non-depositional events within the Cambrian–Ordovician boundary section near Dayangcha, Jilin Province, China, *Geol. Mag.*, 1993, vol. 130, pp. 443–452.

Rodionov, V.P. and Gurevich, E.L., Key magnetostratigraphic sequence of the Lower Ordovician deposits, northwestern Siberian Platform, *Neftegaz. Geol. Teor. Prakt.*, 2010, vol. 5, no. 3. http://www.ngtp.ru/rub/2/38_2010.pdf.

Tolmacheva, T. and Abaimova, G., Late Cambrian and Early Ordovician conodonts from the Kulumbe River section, northwest Siberian Platform, *Mem. Assoc. Aust. Palaeontol.*, 2009, vol. 37, pp. 427–451.

Van der Voo, R., *Paleomagnetism of the Atlantic Tethys and Iapetus Oceans*, Cambridge: Cambridge Univ. Press, 1993.

Yang, Z., Otofuji, Y., Sun, Z., and Huang, B., Magnetostratigraphic constraints on the Gondwana origin of North China: Cambrian–Ordovician boundary results, *Geophys. J. Int.*, 2002, vol. 151, pp. 1–10.

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